

COS750 – Instructional Design Document

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Data Structure: Binary Search Tree (BST)
Algorithms: BST Search & BST Insertion

Summary

The **Binary Search Tree (BST)** is the chosen data structure for the app, and the algorithms implemented are **BST Search and BST Insertion**. A BST is a hierarchical structure where each node has at most two children: values smaller than the parent are placed on the left, and values larger are placed on the right. This property makes searching efficient, as the algorithm systematically compares the target value with nodes and decides whether to move left or right until the value is found or determined to be absent. Similarly, insertion follows the same rule, placing new nodes in their correct position to maintain the BST property. By focusing on these two algorithms, the app introduces students to both the fundamental structure of trees and the logical process of navigating and modifying them, ensuring they build a strong foundation in data structures before tackling more advanced topics.

1. Learning Theories Applied

Constructivism – Building on Prior Knowledge

- Students build their own BSTs by inserting values step by step.
- The BST is introduced as "binary search turned into a structure," linking to prior knowledge of arrays and comparisons.
- Activities encourage prediction before explanation (e.g., "Where does 42 go?").
- Learners discover the ordering rule themselves before it is formally stated.

Cognitivism / Information Processing

- Content is chunked into short lessons (3–5 minutes each).
- Dual coding: every rule is shown as text and an animated tree.
- Worked examples precede practice, reducing cognitive load.
- Signalling: current node highlighted, comparison shown alongside pseudocode.

Behaviourism – Feedback and Reinforcement

- Immediate feedback: green highlight for correct, red with explanation for incorrect.
- Points, streaks, and badges reward progress.
- Repeated practice until mastery (e.g., three correct inserts in a row).

Social Cognitive Theory

- "Watch me" mode demonstrates search/insertion step by step.
- Students observe worked examples, then practice with guidance.
- Progress tracking (badges, progress bar) builds self-efficacy.
- Confidence grows through small successes and unlimited retries.

2. Catering to Different Learning Styles

Fleming's VARK Model

- Visual → Animated BST diagrams, colour-coded paths.
- Auditory → Optional narrated explanations.
- Read/Write → Pseudocode, text notes, downloadable summary sheet.
- Kinesthetic → Drag-and-drop node placement, interactive sandbox mode.

Kolb's Learning Cycle & Honey & Mumford Styles

- Concrete Experience (Activist) → Insert keys freely before rules are given.
- Reflective Observation (Reflector) → Prompts like "What do you notice about small vs large numbers?"
- Abstract Conceptualisation (Theorist) → Formal BST property and pseudocode revealed.
- Active Experimentation (Pragmatist) → Real-world tasks (e.g., build a BST of student IDs).

3. Motivation Strategies

Flow Theory

- Start simple (3-node BST), then scale up difficulty.
- Clear goals and immediate feedback maintain engagement.
- Adaptive difficulty ensures balance between challenge and skill.

Malone & Lepper's Framework

- Challenge → Levels increase in complexity; trick cases like invalid BSTs.
- Curiosity → Prompts like "What happens if we insert numbers in sorted order?"
- Control → Students choose pace, retry options, and sandbox values.
- Fantasy → Themed scenario (e.g., "Forest of Keys") to make learning engaging.
- Recognition → Badges such as Tree Builder, Search Master, Traversal Explorer.

Keller's ARCS Model

- Attention → Opening puzzle, animations, varied activities.
- Relevance → Links to real-world uses (databases, file systems).
- Confidence → Clear objectives, hints, unlimited retries.
- Satisfaction → XP, badges, progress bar, and a "Lecture-ready" certificate.

4. Assessment and Scaffolding

Diagnostic Check (Pre-Lesson)

- Short pre-test on comparisons, binary search, and linked-list nodes.
- Weak students shown refresher content before continuing.

In-Lesson Checks

- After each lesson, 2–3 questions with instant feedback.
- Two wrong answers unlock hints and worked examples.

Final Quiz

- 8–10 mixed questions tagged to Bloom's levels:
  - Remember → "Which node is the root?"
  - Understand → "This tree is a BST. True or false?"
  - Apply → Drag-and-drop insert exercise.
  - Analyse → Compare two trees and explain why one is taller.

Scaffolding Rules

- 80%+ → Lecture-ready, certificate awarded.
- 50–79% → Review weak topics, short re-test.
- Below 50% → Guided walkthrough with hints, then re-test.

5. User Journey

Screen 1: *Welcome*

- Hook: "How does your phone find a contact among 1000 in a split second?"
- Learning objectives listed, reassurance: "Nothing is marked, retries are free."
- Purpose: grab attention, lower anxiety, set expectations.

Screen 2: *Mini-lesson 1, "What is a BST?" (I do)*

- Animated tree with the rule: smaller goes left, larger goes right.
- Options: listen, read, or watch the animation.
- Purpose: introduce the concept simply.

Screen 3: *Worked Example ("Watch me")*

- App inserts 42 step by step, highlighting comparisons.
- Student controls pace with "Next step."
- Purpose: demonstrate algorithm clearly.

Screen 4: *Guided Practice ("You decide")*

- Student predicts each step: Go left, Go right, Found it, or Not in tree.
- Wrong moves explained, not penalised.
- Purpose: scaffold learning through prediction.

Screen 5: *Independent Practice*

- 3–5 questions without hints, increasing difficulty.
- Streak counter visible for motivation.
- Purpose: build confidence through mastery.

Screen 6: *Sandbox Mode*

- Students insert their own numbers and watch the tree grow.
- Curiosity prompt: “Try inserting 1, 2, 3, 4, 5. What happens?”
- Purpose: encourage exploration and discovery.

Screen 7: *Final Quiz*

- 8–10 mixed questions (MCQ, drag-and-drop, trace path).
- Purpose: assess readiness before lecture.

Screen 8: *Results & Scaffolding*

- Score shown with three routes: ready, review weak topics, or guided re-walkthrough.
- Improvement over last attempt highlighted.
- Purpose: diagnose and support, not grade.

Screen 9: *Progress Dashboard*

- Visible progress bar, badges (Explorer, Pathfinder, Key Master, Comeback).
- Purpose: recognition and motivation without competitive leaderboards.

6. App Features Summary

1. Interactive BST Builder → Insert/delete nodes, immediate feedback.
2. Traversal Simulator → Animations, drag-and-drop exercises.
3. Adaptive Quizzes → Scaled difficulty, timed challenges.
4. Gamification Layer → Badges, XP, progress bar.
5. Progress Dashboard → Tracks completed tasks, shows improvement.